

Reproducing Mixup and Investigating Its Effectiveness Under Increasing Label Noise

Paper title: Mixup: Beyond Empirical Risk Minimization — ICLR 2018

ICLR official paper page : https://iclr.cc/virtual/2018/poster/177?utm_source=chatgpt.com

Abstract

This project reproduces the Mixup training strategy introduced by Zhang et al. at ICLR 2018 and investigates how its effectiveness changes under increasing levels of training-label noise. Mixup trains neural networks using convex combinations of pairs of training examples and their labels, providing a simple form of regularization that has been shown to improve generalization. I implemented a simplified Mixup training pipeline using a convolutional neural network on CIFAR-10 and compared it with standard training. On clean data, Mixup achieved 77.17% test accuracy compared with 74.58% for standard training, corresponding to a 2.59 percentage-point improvement. Then I extended the experiment by corrupting 10%, 20%, and 30% of the training labels while keeping the test set clean. The Mixup advantage decreased from 0.99 percentage points at 10% noise to 0.74 percentage points at 20% and approximately zero at 30%. These results suggest that although Mixup improved generalization in our clean-data reproduction, its relative advantage diminished as label noise increased. The experiment demonstrates both the usefulness and limitations of Mixup under imperfect supervision.

Main reproduction question:

“Can we reproduce the generalization benefit of Mixup over standard empirical risk minimization on CIFAR-10?”

Extension question:

“How does the relative effectiveness of Mixup change as the amount of training-label noise increases?”

Introduction

Deep neural networks can achieve strong performance on image classification tasks but may overfit their training data and memorize undesirable patterns. Mixup, introduced by Zhang et al. at ICLR 2018, addresses this problem through a simple training strategy in which pairs of examples and their labels are linearly interpolated. The original work reports improved generalization across multiple datasets and also investigates robustness to corrupted labels.

Problem

The project investigates whether the generalization benefit of Mixup can be observed in a simplified CIFAR-10 experiment and whether this benefit changes when training labels become increasingly unreliable.

Goals

1. Understand and implement Mixup.
2. Compare Mixup against standard training.
3. Reproduce the qualitative generalization finding.
4. Introduce controlled label noise.

5. Measure how the Mixup advantage changes with noise.
6. Analyze limitations and possible explanations.

Dataset

CIFAR-10 : https://www.cs.toronto.edu/~kriz/cifar.html?utm_source=chatgpt.com

CIFAR-10 contains:

- 60,000 color images
- 10 classes
- 50,000 training images
- 10,000 test images
- 32 × 32 pixel resolution

We used the training set for model fitting and retained the test set with clean labels for final evaluation.

The dataset was loaded using **torchvision**.

Baseline

The baseline uses: **Standard empirical risk minimization**

The CNN receives normal CIFAR-10 images and their original labels.

Mixup

For two training examples:

$$(x_i, y_i), (x_j, y_j)$$

we generate:

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j$$

and:

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j$$

where:

$$\lambda \in [0, 1].$$

This is the central Mixup mechanism described in the original paper.

Extension

I independently corrupted a percentage of the **training labels**: 0% 10% 20% 30%

The test labels remained clean.

For every noise level, we compared:

Standard training **vs** Mixup

Results

Label Noise	Standard	Mixup	Mixup Advantage
0%	74.58%	77.17%	+2.59 pp
10%	68.61%	69.60%	+0.99%
20%	71.04%	71.78%	+0.74 pp
30%	69.54%	69.49%	-0.05 pp

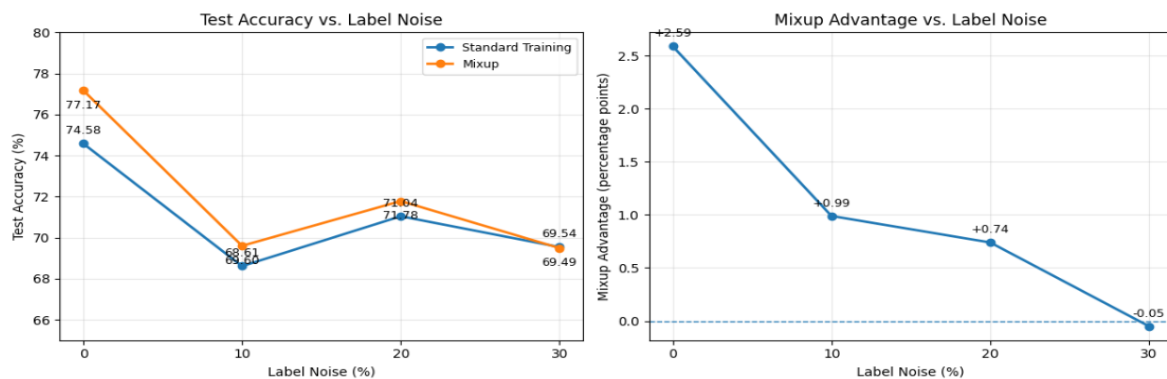


Figure 1. Test accuracy and Mixup advantage under increasing levels of training-label noise. Mixup provides its largest advantage on clean data, while the relative benefit decreases as label noise increases.

Discussion:

On clean data, Mixup improved test accuracy by 2.59 percentage points compared with standard training, providing qualitative support for the generalization benefit reported in the original paper. However, the extension revealed that this advantage did not increase with label noise. Instead, the advantage decreased to 0.99 percentage points at 10% noise and 0.74 percentage points at 20% noise, while the two methods were essentially tied at 30% noise. These results suggest that Mixup's benefit in our experimental setting is strongest when labels are reliable and may diminish when the supervision itself becomes increasingly corrupted.

The accuracy values were not strictly monotonic with increasing noise. For example, standard training achieved 68.61% at 10% noise and 71.04% at 20% noise. This may reflect stochastic variation from model initialization, minibatch ordering, and the particular randomly corrupted

labels used in each experiment. Because each condition was evaluated using a single run, these results should be interpreted as preliminary rather than **statistically conclusive**.

Limitations

1. Single-run experiments

I did not repeat every experiment with multiple random seeds.

2. Simplified architecture

CNN was substantially smaller than the architectures used in the original paper.

3. Limited dataset

I evaluated only CIFAR-10 rather than the multiple datasets studied in the paper.

4. Limited noise models

I used randomly corrupted labels. Real-world annotation errors may have structured or systematic patterns.

5. Computational constraints

The project was intentionally scoped to a small experiment that could be completed within limited computational resources.

Reflection on Learning

The most rewarding part of this project was moving from reading a research paper to actually implementing one of its ideas. Initially, concepts such as paper study, DataLoaders, training loops, and Mixup were unfamiliar, but implementing each component helped me understand how the different parts of an ML pipeline connect. One particularly useful learning experience was understanding that the original CIFAR-10 dataset should remain unchanged while separate noisy versions of the training labels can be created for controlled experiments.

One of the main challenges was interpreting experimental results correctly. Initially, I expected Mixup to consistently outperform the baseline, but the results changed as label noise increased. This taught me that a research experiment is not about forcing the expected outcome; unexpected results can be valuable if they are analyzed carefully. I also learned the importance of distinguishing between a paper's original findings and an extension that asks a new question.

The most important methodological lesson was the importance of controlled comparisons. To evaluate Mixup fairly, the same dataset, architecture, optimizer, and evaluation procedure should be used while changing the training strategy. I also learned that a single experimental run is not enough to make strong statistical claims, which is an important limitation of this project.